

Optimization of Aircraft Runway Scheduling Based on Metaheuristic Algorithms

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1. Work Completed and Progress to Date

Since the project's inception, work has focused on establishing a robust data foundation and selecting the most effective algorithmic framework for the optimization model.

Data Preprocessing and Cleaning: I have initiated the processing of the Brazilian National Civil Aviation Agency dataset, which contains approximately 2.54 million flight records. This involved filtering key variables such as planned/actual departure times, flight status, and airport coordinates.

Statistical Modeling: To derive realistic "hard constraints" for the scheduling engine, I have been applying Poisson and Exponential distributions to the cleaned historical data. The goal is to mathematically determine average airport throughput and safe takeoff/landing intervals.

Algorithmic Research and Selection: Following a comprehensive literature review of Mixed-Integer Linear Programming (MILP) and metaheuristics, I have selected Ant Colony Optimization (ACO) as the primary metaheuristic framework. ACO was chosen for its proven ability to handle wake vortex constraints and minimize both total delay and inter-airline unfairness more effectively than traditional First-Come-First-Served (FCFS) methods.

System Architecture Design: I have outlined the full-stack application logic, where the backend will handle complex optimization calculations while the frontend GUI provides a visualized, user-friendly interface for inputting flight and weather parameters.

2. Identified Problems and Challenges

Despite the progress in algorithm selection, the following technical challenges have emerged during the data processing phase:

Data Integrity and Volatility: During the cleaning process, a significant volume of "anomalous" data was identified. The initial aggressive filtering resulted in the removal of a large portion of the dataset.

Statistical Inaccuracy: The loss of data has led to potential inaccuracies in the Poisson and Exponential distribution calculations. These inaccuracies make it difficult to establish reliable baseline values for airport throughput and safety intervals, which are critical for the optimization model's constraints.

Complexity of Multi-Objective Balancing: Refining the weighted optimization function to maximize runway utilization while simultaneously ensuring scheduling fairness and delay minimization remains a complex modeling task.

3. Future Work Plan and Schedule

To address the current data issues and move toward system implementation, the following steps are planned for the next move:

Refinement of Data Cleaning (March): Instead of simple deletion, I will implement more sophisticated outlier detection and data imputation techniques. This will ensure that the statistical modeling of throughput and safety intervals is based on a more representative and accurate subset of the 2.5 million records.

Algorithm Development and Coding (March – April): I will begin the core implementation of the ACO algorithm. This includes defining the pheromone update rules and heuristic factors specifically for multi-runway scheduling.

System Integration and GUI Construction (April): Once the optimization engine is stable, I will develop the frontend GUI. This will allow for the visualization of optimized timetables and a comparison of predicted delay reductions against historical actuals.

Testing and Robustness Evaluation (Late April): The final stage will involve an 80/20 train-test split of the Brazilian data. I will also try to introduce "weather noise" simulations to verify that the generated schedules remain reliable under uncertain real-world conditions.